LAB4: MICROSTRIP ANTENNA

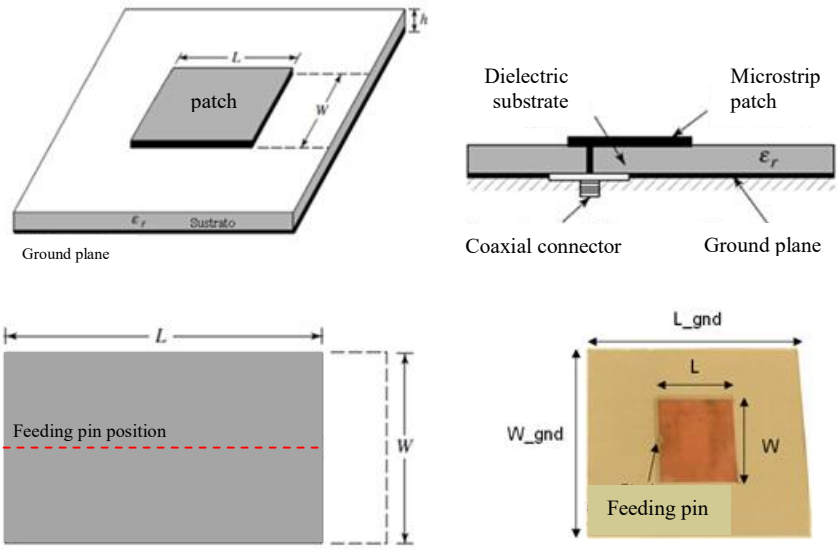
1. Objectives and preliminary considerations

The main goal of this laboratory session is to simulate a microstrip antenna, analyse the results and adjust the structure to the requested operating frequency: 900 MHz.

In section 2 there are some questions you can solve at home before starting the laboratory session.

The deadline for delivering the report is May 25th. Please, use a one column format for the wording, answer each question one by one, and write your name in the file name.

Prior to the start of the lab you can download the document Altair Feko patch antenna.pdf from Campus Virtual.



2. Previous exercise

Q1. Use the equations from the slides to find dimensions *W* and *L* for a microstrip patch antenna operating at 900 MHz, use a dielectric substrate with thickness h = 1.55 mm, permittivity εr = 4.5 and loss tangent tan δ = 0.02.

Q2. <u>After a reading of the document Altair Feko patch antenna.pdf, list all steps to create and launch the simulation.</u>

Q3. Define relative dielectric permittivity and loss tangent? How do they affect to the patch antenna?

3. Practical session

3.1. Definition of variables

Create the following variables used to define the structure.

Name	Expression	Comment (you needn't to copy them)
freq_min	0.6e9	Min frequency of simulation (Hz)
freq_max	1.2e9	Max frequency of simulation (Hz)
lambda_min	c0/freq_max*1000	Wavelength (mm)
patch_L	Calculated in Q1	Patch length (mm)
patch_W	Calculated in Q1	Patch width (mm)
substrate_er	4.5	Dielectric permittivity
substrate_h	1.55	Dielectric thickness (mm)
substrate_tand	0.02	Dielectric tangent loss
feed_dist	16	Feeding pin position from the edge (mm)
feed_rad	1.3/2	Pin radius (mm)
substrate_W	160	Substrate ground width (mm)
substrate_L	100	Substrate ground length (mm)

3.2. Infinite substrate model

Follow the chapter 3 of the Getstarted.pdf document to create the patch antenna model with the variables from the chart, launch the simulation and see Postfeko results.

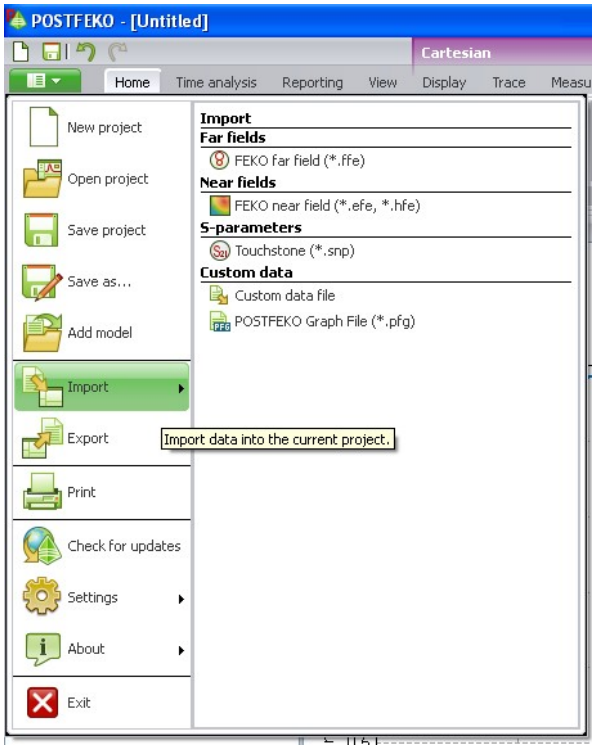
Q4. Shown the impedance (real and imag) and the reflection coefficient. Find the best matching point and the relative bandwidth at -10 dB. Comment on the result.

Q5. Run a wide band frequency simulation from 500 MHz up to 2.5 GHz and identify the rest of operating modes. Move the feeding point to the middle of the L side. Compare the results on the same graph and explain the differences

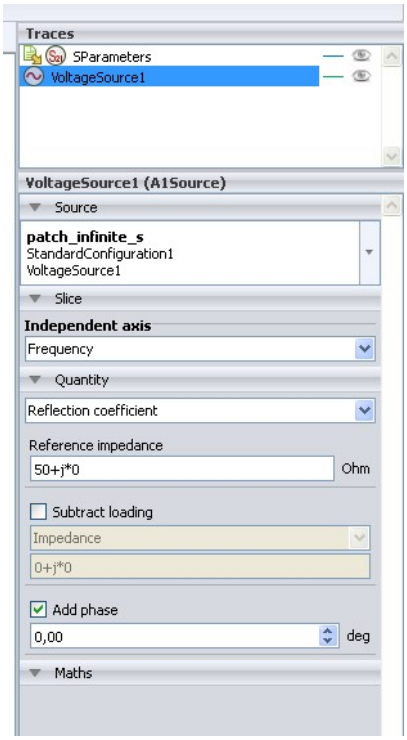
Q6. Plot the matching for a 100 Ω impedance. Achieve the best matching as possible for a 100 Ω impedance. What do you have to do? Why? Show the currents on the antenna.

Now, introduce in Postfeko the result from the measurement to compare with the simulation (Import >> Touchstone, select the file patch900_s.s1p available in Campus Virtual, see next picture). Once data are imported, compare them with the results of the first simulation of Q5 (using a reference of 50 Ω).

Q7. What is the relative error in frequency made by the simulation for the fundamental mode? Next, compare both results in the Smith chart. Comment the differences. In the simulation result use the *Add phase* available in the right menu to achieve the same line for both results, see next picture.



Import touchstone file.



Add phase.

3.3. Finite substrate model

Follow the chapter 3 of the Getstarted.pdf document to create the patch antenna model with the variables from the chart, launch the simulation and see Postfeko results.

Q8. Show the model with finite dielectric. Compare the new results with the infinite substrate model. Which are the main differences and similarities?

Q9. Show the efficiency and comment the value. At the frequency of maximum matching, show the radiation pattern in 3D, the E-plane cut and the H-plane cut for the main polarization.

Q10. Imaging that now we want to use the patch for a communication system in the ISM 868 MHz band. Tune the patch dimensions and feeding point location for optimal performance (maximum impedance matching) in the 868 MHz band. Explain the patch adjustment. Show the matching plot and the final dimensions for the 868 MHz patch